THE GRAMMAR OF A FIRE ALARM SYSTEM

How Buildings Tell Us What Is Happening



RESCUELOGIC®

A RescueLogic Publication

Dan Horon
President

THE GRAMMAR OF A FIRE ALARM SYSTEM

How Buildings Tell Us What Is Happening

PANEL_ID • DEVICE_ID • DEVICE_TYPE • DEVICE_LOCATION • STATUS

RESCUELOGIC®
A RescueLogic Publication

Dan Horon
President

Publication Information

The Grammar of a Fire Alarm System

How Buildings Tell Us What Is Happening

Dan Horon, President

A RescueLogic Publication

Cadgraphics Incorporated

Safety Made Simple

This guide is intended as an educational introduction to fire alarm information. It does not replace applicable codes, manufacturer instructions, approved plans, testing, or site procedures.

SAFER is a proposed format for sharing structured fire alarm information between systems. It is included here as an example of how the five basic categories can be carried in a common form for discussion, testing, and refinement.

RescueLogic® is a registered trademark of Cadgraphics Incorporated.

Author's Note

This is a short guide to the information inside a fire alarm system, and how to use that information to identify what happened, which device reported it, what kind of thing it is, where it is, and which system is speaking.

The five-category model described here is a teaching tool based on information commonly found in fire alarm systems.

Chapter 7 briefly discusses SAFER, a proposed shared reporting form built from these basic categories.

The goal is simple: make important fire alarm information easier for people and computers to understand.

Contents

Publication Information 2
Author's Note 3
Introduction — How Buildings Communicate 5
Five Categories at a Glance 6
1. Every Alarm Is a Sentence 7
2. Something Changed 9
3. Which One? Where Is It? 11
4. Different Dialects, Same Meaning 13
5. Same Information, Different Views 14
6. When Information Does Not Match 16
7. Sharing the Same Information 18
8. Take Action 19
Conclusion — Keep the Information Clear 21
Quick Reference 22
Glossary & Source Notes 23
About the Author & Resources 24

Introduction — How Buildings Communicate

In 1986, the company I worked for was bidding on a project that called for a computer to display alarms on a floor plan.

I thought: **“AutoCAD could do that.”**

I had spent two years using AutoCAD and writing code to make fire alarm drawings faster and easier to read. Alarms on floor plans felt like a natural next step. A floor plan could show the building as people knew it. A detector could appear in the room where it was installed. The computer could zoom from a whole floor to one device.

I thought I was making better fire alarm graphics. Over time, I realized I was making fire alarm information easier to understand.

Three Questions

Our first campus-wide integration was at 3M Headquarters.

The system was new to all of us, and everyone wanted to get it right. The guards practiced with it for about six months. It monitored fire, security, mechanical, and environmental alarms. When something confused them, I’d make it better.

I wanted a new guard to be able to look at the screen and answer:

What happened? • Where did it happen? • What should I do?

Those questions became my test for every change: Does it help answer them?

I also wanted the system to stay useful without needing me to keep it updated. I used tools each trade already knew. Technicians and inspectors could update device information. IT and facility staff could add floor plans. Authorized users needed to be able to move device symbols, change labels, and improve locations as the building changed.

One other thing was obvious: **ACTIVATED and TROUBLE are not the same.**

Fire alarm systems watch themselves for certain problems. This is called **supervision**. A low battery, missing device, or failed communication path can create TROUBLE.

TROUBLE means the system found a problem that may need service. It does not mean that a detector or pull station has activated. ACTIVATED tells the operator that something being monitored has operated. The system should make that difference clear without needing an installer there to interpret it.

The rest of this guide organizes fire alarm information into five basic categories. They give people and computers a simple way to answer the basic questions.

Five Categories at a Glance

The Sentence Underneath the Alarm

Imagine one smoke detector. Its identifier is L1S067. It is in Classroom 4 West. It belongs to a panel we will call Panel 1. Then it activates.

Put those facts together:

> **Panel 1 reports that smoke detector L1S067 in Classroom 4 West has activated.**

The technology behind the sentence may be complicated, but the sentence structure is simple and consistent.

Field	Question	Example
PANEL_ID	Who is reporting?	PANEL 01
DEVICE_ID	Which thing?	L1S067
DEVICE_TYPE	What kind?	Smoke Detector
DEVICE_LOCATION	Where?	Classroom 4 West
STATUS	What is reported?	1 (Active)

> **The field names are technical. The questions are ordinary.**

These five categories summarize the basic information people need. A complete SAFER v1.1 event message, however, has seven required fields, including a numeric STATUS reporting code. The optional CONDITION field can preserve more descriptive information about what the source reported.

Once those relationships are clear, the same information can appear in a list, on a floor plan, in a report, in a database, or in a message sent to another computer.

Throughout this guide, words such as ACTIVATED and TROUBLE describe conditions in familiar human language. SAFER uses a separate numeric STATUS field. For example, SAFER STATUS=1 means Active, and SAFER STATUS=3 means Restore. The SAFER Protocol defines the transmitted values.

PART I — LEARN THE LANGUAGE

1. Every Alarm Is a Sentence

Start With Meaning

A fire alarm sounds. A display changes. The first useful question is not which protocol carried the message or how the circuit was wired.

It is: What happened?

Then: Where?

A real system needs enough precision to identify the thing. That is why the smoke-detector sentence needs more than “a detector activated.” The system needs to know the reporting system, the exact point, the kind of point, its location, and its present condition.

Identity, Meaning, Place

An electrical identifier tells the system which device. A location tells a person where to go. Alarm systems are designed to tie those facts to a single point.

L1S067 may be the exact identity a technician needs. Classroom 4 West is the destination a responder needs. A third fact, that the device is called a Smoke Detector, conveys what L1S067 represents.

The system needs a unique identity. A person needs to know what to look for and where.

Those facts are worth keeping separate. The computer can combine structured facts in a predictable sentence. It is much harder to recover clean facts after everything has been mashed into one short display description.

The Same Categories Still Apply

Here are three different events. Each could require a very different response, and even different crews. Yet the same categories describe them consistently.

Events Reported from PANEL 01		
Device (Type & ID)	**Location**	**Condition**
Pull Station 1M001	Conference Hall Exit	ACTIVATED
Waterflow L1S002	Sprinkler Room	ACTIVATED
Smoke Detector L1S067	Classroom 4 West	TROUBLE

The device changes. The condition changes. The five categories still apply.

That is useful for people because it creates a familiar pattern and helps direct the right response. For computers, every fact has a clear and precise role.

The Same Information Can Appear in Different Ways

A list, a floor plan, a printer message, a database record, and a JSON object can all describe the same point.

They do not have to look alike, but they should not mistakenly disagree.

If the list, floor plan, and database describe the same device, they should agree.

2. Something Changed

Before an alarm is an alarm, something happened.

Smoke reached a detector. Water began moving through a sprinkler pipe. A valve moved from its expected position. A communication path stopped working.

ALARM, SUPERVISORY, and TROUBLE are different classifications. The underlying change comes first.

First understand what changed. Then describe what that change means.

Activated Does Not Mean Alarm

A smoke detector, waterflow switch, or valve-monitoring switch can activate. Those activations do not necessarily receive the same classification or response.

What happened	Reported point condition	Typical classification
Smoke detector senses smoke	ACTIVATED	Fire Alarm
Waterflow switch operates	ACTIVATED	Fire Alarm
Valve supervisory switch operates	ACTIVATED	Supervisory
Communication path fails	TROUBLE	Trouble

ACTIVATED describes the point condition. Fire Alarm adds classification for handling according to plan.

An activated point tells us something has operated. A trouble tells us the system has found a problem that needs attention.

Sometimes the same word can describe both the condition and its classification, as with TROUBLE.

A Simple State, a Meaningful Description

People can recognize a few familiar states while receiving many different descriptions. RescueLogic uses Normal, Alarm, and Trouble as three Base States for device presentation. A Trouble report may describe Open Circuit, Ground Fault, or Detector Missing.

The state helps people recognize the situation. The descriptive condition helps explain what the source reported. SAFER's six numeric STATUS codes are distinct from RescueLogic's three Base States.

Words can show what happened, while labels and visual cues can show how it was classified. For many people, color can make those differences easier to notice.

State Is Not an Event

A state describes a condition at a particular time. An event records a change or observation. A report tells us what the source communicated. They are related, but not the same thing.

Fire alarm panels supervise their conditions and can report a return to Normal. A receiving application may have less information; silence alone does not establish Normal. A report can also describe an earlier condition that is still present.

At 10:14, P1 changed from NORMAL to ACTIVATED, and the event was classified as a fire alarm.

At 10:42, P1 changed back from ACTIVATED to NORMAL.

The first event records the activation. The second records the restoration. Together they create the history of the point.

3. Which One? Where Is It?

A Name Is Not a Place

People already understand this distinction. The name on an envelope shows the recipient. The address underneath tells where to deliver it.

A fire alarm point works the same way.

DEVICE_ID answers: Which one?

DEVICE_TYPE answers: What kind of thing?

DEVICE_LOCATION answers: Where is the device located?

An address such as N54L32D001 may be perfectly useful to a technician and nearly useless as directions. The exact address can remain available while a clearer human location is added.

Location for Someone Who Doesn't Know the Building

"Hallway" may be technically correct but gives little guidance in a large building.

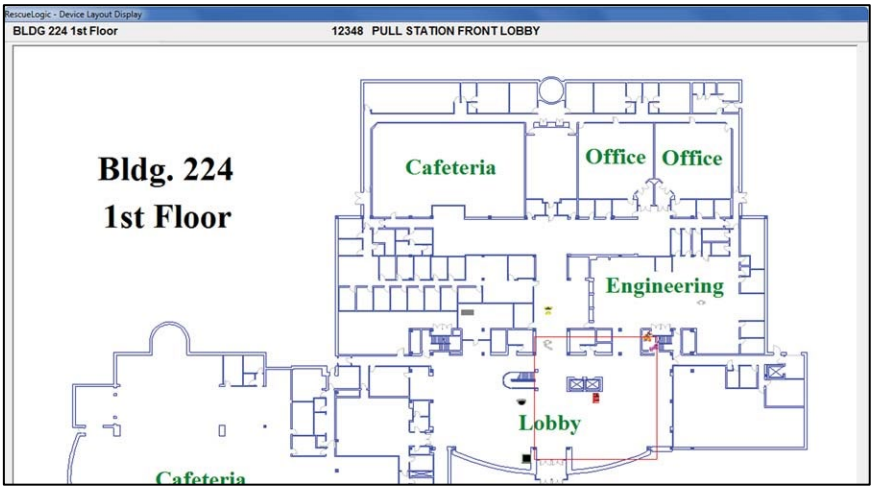
"Third Floor East Corridor Outside Room 312" gives a person a destination.

The programmed location in the source system should remain available. A clearer owner-facing description can be added without changing the source wording.

The source location can remain available while a clearer human location is added.

A Floor Plan Adds Orientation

Text can name a destination. A floor plan can show the destination in relation to doors, corridors, stairs, and nearby rooms.



Coordinates and DEVICE_LOCATION are different kinds of information. Coordinates tell software where to draw. Location text tells a person where to find the device.

Location names the destination. The floor plan provides orientation.

If the written location says Classroom 4 West but the symbol is drawn in Classroom 5 East, the computer does not need to know which source is correct to know that the two descriptions disagree.

PART II — SAFETY MADE SIMPLE

4. Different Dialects, Same Meaning

Different manufacturers use different words for similar information. One system may say point. Another may say module or sensor. Technicians may also encounter terms such as node, loop, card, circuit, input, and output. The common pattern became visible by first understanding how each manufacturer reported information in its own way.

A beginner does not need to master every manufacturer's vocabulary and part numbers before understanding the event.

The five categories describe the basic event. Manufacturer-specific details can be added when needed.

The Practical Idea

The exact manufacturer term may still matter for technical work. At the same time, the basic information has a simple job for other users.

Exact identity The panel and point identifiers still tell us exactly which point.	**Device type** The device type tells us what the point represents.
Clear location The location tells a responder where to go.	**Reported condition** Describes what the source reported, which may include activation, trouble, or restoration.

5. Same Information, Different Views

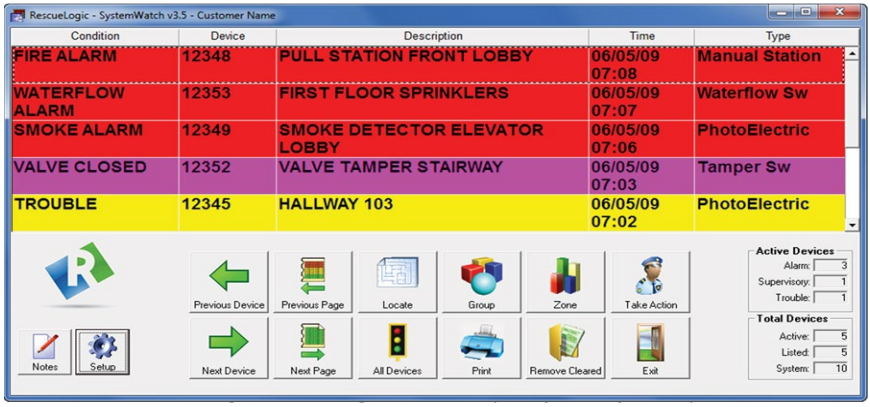
The technician, guard, facility engineer, and firefighter may all be looking at the same event. Each person may see different details from the same event.

Different People Need Different Information

A guard may want to know what happened and where.

A technician may need the exact panel and point address.

A firefighter may need the floor plan.



A floor plan is one view. A report is another. Take Action is another.

Different screens can show different details, but the facts should still match.

Store the Information Once. Show It in Different Ways.

RescueLogic stores device information separately from the screen that displays it. The same device record can appear in a list, on a floor plan, in event history, in a report, or in a Take Action message.

ONE DEVICE RECORD

LIST Active events and time	**FLOOR PLAN** Location and orientation
EVENT HISTORY What changed and when	**REPORT** Documented information
TAKE ACTION Site-specific response information	**TECHNICAL VIEW** Exact panel and point details

Each view has a different job. The underlying device information should stay consistent.

6. When Information Does Not Match

When the same device appears in a list, database, and floor plan, the computer can compare those sources and flag information that does not match.

A Simple Mismatch

Suppose the device record says L1S067 is a Smoke Detector in Classroom 4 West.

The floor plan contains a symbol for the same PANEL_ID and DEVICE_ID, but the symbol is a pull station.

Those descriptions disagree about device type. The difference tells us what needs review, not which record needs correction.

The computer can recognize a discrepancy without deciding which record needs correction.

When the properties of devices are listed in columns, missing information and mismatches become obvious. That is useful because a technician can see what needs review before responders have to depend on it.

Useful Checks

The five categories make several useful checks possible:

Check	Practical question
Missing	Is the device type or location missing?
Duplicate	Do two records use the same panel and point identity?
Type	Does the listed device type agree with its symbol and known configuration?
Location	Does the written location agree with its position on the floor plan?
Live event	Can the reported point be matched to its maintained device record?

A difference invites verification; it does not by itself tell us which record to change. The software should show its work and not silently rewrite a value because another source disagrees.

Comparing Different Manufacturers

Manufacturers may describe the same type of point differently. Before comparing two systems, the software must first understand what each manufacturer's terms mean.

In all cases, the goal is to explain the event in terms the people responding can understand, while keeping the manufacturer's original message available.

The original panel message makes the translation easier to check later.

PART III — SHARE AND RESPOND

7. Sharing the Same Information

Computers can also exchange the five basic categories.

One example is SAFER—the Security And Fire Emergency Reports.

In JSON, the same event can look like this:

```
{
  "PANEL_ID": "PANEL 01",
  "DEVICE_ID": "L1S067",
  "DEVICE_TYPE": "Smoke Detector",
  "DEVICE_LOCATION": "Classroom 4 West",
  "STATUS": 1
}
```

This is an illustration of the five core fields, not a complete SAFER message.



Current SAFER specification

github.com/rescuelogic/SAFER

SAFER is offered for discussion, testing, and refinement.

The important idea is simply that another computer can understand the event more easily when the information is clearly named.

8. Take Action

The five categories help answer two of the three questions that shaped my early systems:

What happened? • Where did it happen?

Corrine Named the User's Question

One of my early buttons was called Device Help. I had thought the name described the feature.

My wife Corrine suggested a better name: Take Action.

She was right. Device Help described the software. Take Action addressed the person using it.

The computer should speak from the user's point of view.

The person asks:

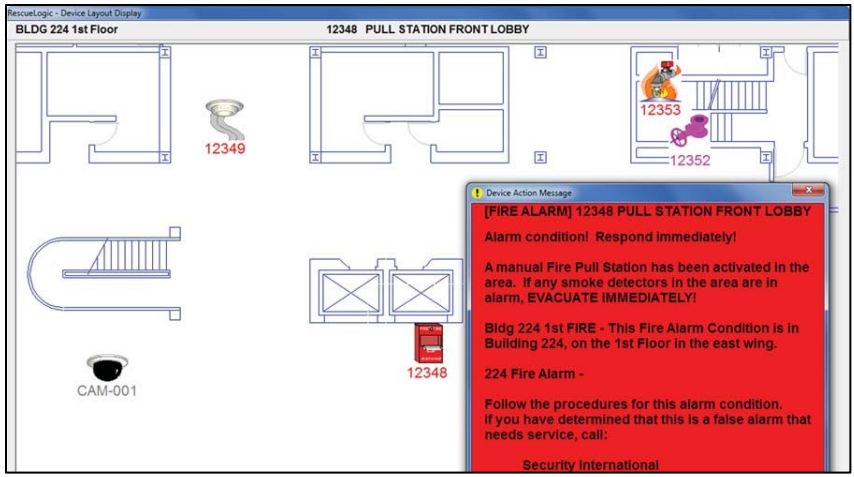
What should I do?

One of the first things to notice is whether the system is reporting ACTIVATED or TROUBLE. Those conditions can lead to very different procedures, depending on the situation.

The software can show approved information. It does not make on-the-spot policy or tactical decisions.

The alarm information describes what is being reported. Response information depends on the person, the site, and approved procedures.

A guard may need a site procedure. A technician may need diagnostics. A firefighter brings authority and training that the building computer does not have.



A RescueLogic Take Action view shows site-specific response information over a floor plan.

When the Responder Does Not Know the Building

Now imagine the fire department arriving at a building the responders have never seen.

The fire alarm system may already contain the panel, point, type, location, condition, and floor-plan position. The responder may know none of it.

The missing step is the handoff.

A printed or mobile floor plan can carry location and orientation to the responder.

POINT	→	**FLOOR PLAN**	→	**RESPONDER**
Exact device		Where it is		Location + orientation

The point address identifies the device. The floor plan helps the responder find it.

Conclusion — Keep the Information Clear

Fire alarm technology will continue to change. The basic information people need is much more stable:

Who is reporting?
Which device?
What kind of device?
Where is it?
What is happening?

A technician may need more technical detail. A guard may need a simple description. A firefighter may need a floor plan. Another computer may need structured data.

The presentation can change. The important thing is that the information remains accurate, consistent, and understandable.

Technology changes. The basic information people need stays remarkably consistent.

Applicable codes and standards set the requirements. The goal is to meet them in the clearest, most useful way for the people whose safety depends on the system.

Quick Reference

Field	Question	Remember
PANEL_ID	Who is reporting?	Context
DEVICE_ID	Which thing?	Preserve exact identity
DEVICE_TYPE	What kind?	Use a useful noun
DEVICE_LOCATION	Where?	Human destination
STATUS	What is reported?	Defined reporting code; not necessarily current physical state

Three Familiar States

These states describe a familiar human presentation, not additional SAFER STATUS codes.

NORMAL	A monitored condition reported or established as normal. Silence alone does not establish Normal.
TROUBLE	A reported problem needs investigation.
ACTIVE	A monitored point has operated; follow applicable procedures.

Three Questions the System Should Help People Answer

What happened? • Where did it happen? • What should I do?

Five Ideas Worth Keeping

- Meaning comes first.
- Original panel information remains useful when translating.
- Location text gives the destination; the floor plan adds orientation.
- Different screens may show different details, but the facts should match.
- Computers can flag information that does not agree.

Glossary & Source Notes

Classification — The category used to identify how a reported condition is handled, such as alarm, supervisory, or trouble.

Event — A change or observation involving a known point. A transmitted report may describe a new change, restoration, a continuing condition, or a status observation.

Supervision — The fire alarm system's way of watching for problems that could affect operation.

Trouble — A condition reported when the system detects a problem that needs attention.

SAFER — Security And Fire Emergency Reports, a reporting format discussed in this guide.

Source — What was actually received from the originating system.

Take Action — The RescueLogic name for site-specific response information associated with an event.

Translation — Interpreting a source message while keeping enough of the original information to check the interpretation.

View — A presentation of shared facts, such as a list, floor plan, report, or technical display.

Source Notes

RescueLogic System Watch: A Handbook for Guards, Dispatchers, and First Responders. Cadgraphics Incorporated, 2014.

RescueLogic Product Manual. Cadgraphics Incorporated, Revision 5.0.

Horon, Dan. The SAFER Protocol: Security and Fire Emergency Reports, Version 1.1 (proposed specification). RescueLogic / Cadgraphics Incorporated, October 2026.

T. Bray, ed. The JavaScript Object Notation (JSON) Data Interchange Format, 2017.

About the Author & Resources

Dan Horon is the founder and president of Cadgraphics Incorporated and the creator of RescueLogic software. He has spent decades designing fire alarm graphics, software, and information systems around three practical questions: What happened? Where did it happen? What should I do?

His work has included fire alarm design, graphical annunciation, software development, and participation in NFPA technical committee work.

A RescueLogic Publication • Safety Made Simple

SAFER

The current SAFER specification and project files are available here:

github.com/rescuelogic/SAFER

Three Questions to Keep

What happened?
Where did it happen?
What should I do?

RescueLogic® is a registered trademark of Cadgraphics Incorporated.

NOTES

NOTES